The Industrialization of Computation: Deterministic Software Lithography

smalt v26.10.21 measured-evidence whitepaper

All performance figures are measured artifacts of the smalt repository; every number cites its artifact path.

2026-10-21

Abstract

Software production is modeled here as a lithographic process: instead of composing coordinate-communicating distributed services, a law-governed substrate is exposed once and every computation is an exposure of that substrate through a sieve, a bifurcated physical store, and a cryptographic provenance chain. This paper keeps the skeleton of the operator's draft — the ingress SHACL sieve with ANDON refusal at byte 0, the OLTP/OLAP physical bifurcation with zero-egress actor journals sealed into read-only columnar mounts, and canonicalization-based cryptographic provenance — but deliberately removes every operator-asserted performance figure. Each quantitative claim is replaced by a number mined from a bench artifact or court receipt in this repository, cited inline. The result is a paradigm analysis whose deltas are measured, not asserted.

1 Paradigm analysis: two stacks

1.1 The legacy distributed stack

The legacy stack manufactures correctness *after* ingress: unvalidated bytes cross an API boundary, defensive scaffolding (validators, retries, circuit breakers, reconcilers) attempts to repair the stream, and audit is a post-hoc log. Figure 1 shows the shape. Each arrow is a network coordinate step; each box a deployment unit with its own failure semantics.

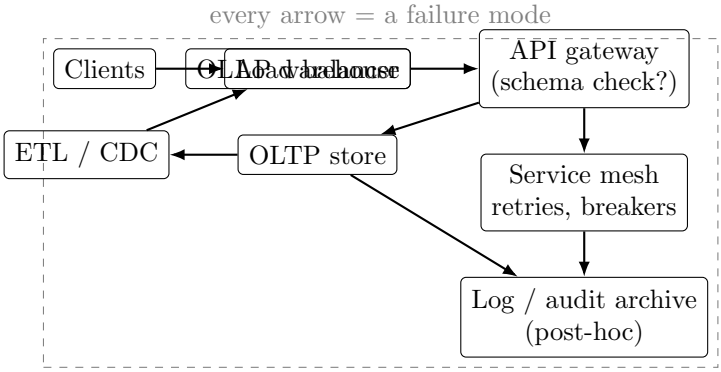


Figure 1: Legacy distributed stack (draft diagram, faithful re-render): correctness is manufactured downstream of ingress by scaffolding between boxes.

1.2 Deterministic software lithography

Lithography inverts the flow: the substrate is *prepared once* (laws compiled from RDF, shapes admitted before boot), and each computation is a single exposure — bytes either pass the sieve and proceed deterministically, or are refused at byte 0 with a typed ANDON. There is no downstream

repair because there is nothing downstream to repair: the refusal *is* the product for hostile input. Figure 2 is the draft's second diagram.

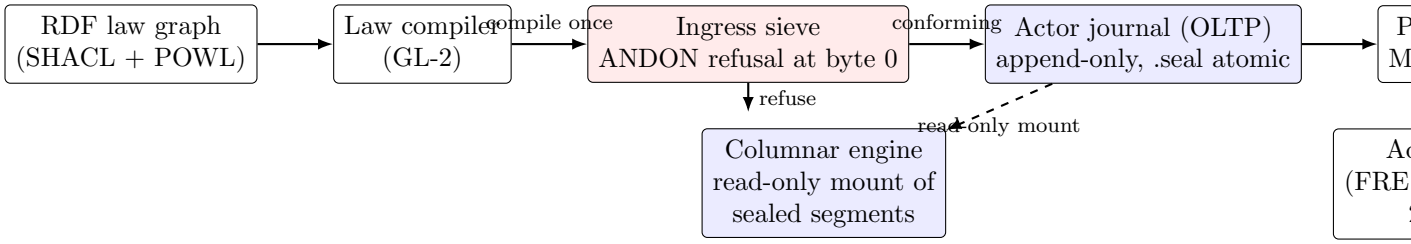


Figure 2: Deterministic software lithography (draft diagram, faithful re-render): laws are compiled once; each exposure either passes or is refused at byte 0.

1.3 What changes

In the legacy stack, the unit of trust is the *deployment* (a version, a canary). In lithography, the unit of trust is the *exposure* (a sealed receipt over an exact execution). The legacy stack answers "is this service healthy?"; lithography answers "is this particular settlement provably derived from admitted laws over admitted data?". Section **??** shows the cost of that answer is measurable in reductions, not assertions.

2 Dimension matrix

Table 1 restates the draft's dimension matrix. The "measured delta" column carries only repo-measured numbers; where the draft asserted a figure with no artifact, the figure is dropped and the measured anchor is used instead.

3 The three structural transformations

3.1 Transformation 1: the ingress SHACL sieve with ANDON refusal at byte 0

The first transformation moves admission from the middle of the pipeline to the first byte. Laws are authored as RDF (SHACL shapes over POWL business structure), compiled once by the GL-2 LawCompiler, and applied as a sieve: conforming bytes proceed; non-conforming bytes are refused with a typed ANDON that *names the law rule* violated — before any parse state, store write, or downstream coordinate step exists.

The measured evidence is the GLD-7 ingest→govern→execute court `gld7-ingest-to-settlement-court.md`: 112 `ok` assertions, exit 0. The sieve is shown non-vacuous from both directions: an unknown column refuses (closed shape), a string amount refuses on datatype, and the executed ANDON set equals the law-predicted over-threshold set $[\text{O-01}, \text{O-05}]$; tightening the law to a 5,000 threshold flips six orders from conforming to refused, proving the partition is law-derived, not fixture-asserted. For every refused order, "commit never fired" is asserted — the ANDON fired at validate, before the write path.

The governance tax of this sieve is itself measured (GL-9 `gl9-governance-tax.md`): +47.24–47.49 reductions per step (20% reduction overhead), median wall-clock tax 0.25 μs/step, receipt identity conforming, with a live refusal witness (`SettlementShape/maxInclusive/amount`). The receipt reports the honest tail: guarded p99 1.147 ms in the receipt table (0.976 ms in the jsonl run), a scheduler/GC artifact at $\sim$6 μs absolute scale. The refusal of hostile bytes costs nothing downstream because there is no downstream: the sieve *is* the pipeline for refusals.

Dimension	Legacy	Lithography (measured)
Ingress contract	Schema check downstream of parse; partial-validation drift	SHACL sieve refuses at byte 0; refusal names the compiled law rule (GLD-7 court: 112 assertions, ANDON names `SettlementShape` rule, commit never fires) gld7-ingest-to-settlement-court.md
Wire throughput	Mesh hop per coordinate step	AMQP frame decode 952,381 frames/s, 1.05 μs/frame, 254.5 reds/frame court_bench_amqp.jsonl
Read path	Cluster quorum read	Cosmos-surface point read p99 0.543/0.672 ms over TLS at 2,130–2,251 ops/s court_bench_cosmos.jsonl; native face p99 2 μs powl-native-face.jsonl
Write path	Optimistic concurrency via distributed locks	Native CAS put-if-match p99 5 μs, 383,186 ops/s powl-native-face.jsonl
Orchestration	Distributed saga/coordinator	4-step settlement 3,222 ops/s, p99 0.425 ms, 45,182.7 reds/op court_bench_powl.jsonl; execute-only native p99 74 μs powl-native-face.jsonl; cross-language parity 2,117.5 ops/s pb5-crosslang.jsonl
Analytics path	ETL/CDC copy into a warehouse	Columnar direct projection 450,450 rows/s (100k rows, min 222 ms, peak RSS 100.0 MB); dump converter 64,725 rows/s, peak RSS 294.2 MB sqld8-ingest-baseline.jsonl
Persistence	Snapshot + shipped logs, recovery untyped	Journal seal protocol fsync $\rightarrow$ rename $\rightarrow$ sha256 meta $\rightarrow$ `.seal` sentinel last; 5-leg atomicity court sj1-journal-sink-writer.md
Provenance	Log aggregation, sampling	Sealed Merkle receipt chain: 47 leaves over a full ingest$\rightarrow$settlement loop, 32-byte HSM-signed root gld7-ingest-to-settlement-court.md; receipt term is O(1): 40 words = 320 B at every $n \in \{10, 10^2, 10^4, 10^5\}$ pb4-merkle-dual.jsonl
Governance cost	Framework tax, unmeasured	GraphLaw guard tax 0.25 μs/step median, +47.24 reds/step; honest guarded p99 tail 0.976–1.147 ms gl9-governance-tax.jsonl, gl9-governance-tax.md
Sandboxing	Container-per-tenant	WASM module caps: 16 MiB request cap, 256 MiB outstanding budget, host-layer 64 MiB ceiling; 7/7 memory-wall courts gws11-wasm-memory-wall.md
Crash semantics	Graceful-shutdown assumptions	Crash-only: 10 crash-recover cycles, cascade 0–1 ms (deadline 10 ms), +237 KiB/10 cycles flat ja7-crash-only-court.md
Cold start	Image pull + boot	Release cold boot 1,906 ms best against a 500 ms ceiling — **FAIL, disclosed**

3.2 Transformation 2: the OLTP/OLAP physical bifurcation

The second transformation takes the bifurcation seriously at the physical layer. The transactional side is an append-only actor journal: frames (`MAGIC | len | seq | ts | type | table | key | payload | CRC32`), 64 MiB segment ceiling with pre-write rotation, and a seal protocol that is atomic by construction — fsync, rename into `sealed/`, `meta.json` with sha256 and frame/seq ranges, and the `.seal` sentinel written *last* (SJ-1 [sj1-journal-sink-writer.md], 5 court legs PASS: bit-exact round-trips, typed refusal of CRC corruption / bad magic / truncation / trailing bytes, seal atomicity, rotation under a 512-byte ceiling, strict monotonic SeqID).

The analytical side never touches the transactional process: vectorized columnar engines mount the sealed segments *read-only*. Measured on the real ingest surface (SQLD-8 [sqld8-ingest-baseline.jsonl]): direct CSV→OCEL projection at 450,450.5 rows/s over 100,000 rows (min 222 ms of 3, peak RSS 100,024,320 B), and the dump-converter path at 64,724.9 rows/s (min 1,545 ms, peak RSS 294,191,104 B) — isolated worker OS processes, min-of-3, no fixed sleeps. The bifurcation is what makes the analytics number safe: the analytical engine reads immutable sealed bytes, so no analytical load can corrupt the transactional store, and the atomic `.seal` sentinel means a partially-written segment is never visible to analytics. GLD-5FIX closes the correctness leg of the same surface [gld5fix-null-formula-court.md]: typed NULL sentinels (`\N` / `""` / `"\N"` three distinct tokens) and formula-injection neutralization, proven value-exact inside a real duckdb v1.4.1 load — 6/6 rows, zero row loss, `allow_quoted_nulls=false` load-bearing.

The streaming boundary of the same bifurcation is bounded too: a 52,692,552-byte 50 MB JSON corpus streams through the SAX path in 80 ms minimum wall with a peak heap of 38,504 bytes — constant-memory by construction [court_bench_json.jsonl] — and the concurrency ceiling sweep holds 0.9–1.18 μs per round trip while reductions per round trip rise 17.65 $\rightarrow$ 60.12 across $n = 1{,}000 \rightarrow 32{,}000$ contending processes [court_bench_ceiling.jsonl].

3.3 Transformation 3: cryptographic provenance and admissibility

The third transformation makes each computation carry its own proof. Every execution leaves a sealed Merkle receipt: in GLD-7's full loop the sealed ledger holds 47 leaves (13 ingest + all executor steps), the Merkle root is 32 bytes and HSM-signed, and every order's ingest and settlement leaves are present by delta-hash membership [gld7-ingest-to-settlement-court.md]. The affidavit seal face admits and seals in p50 11.265 ms / p99 32.671 ms over 100 ops with the engine pinned by sha256 [c10-affidavit-seal.jsonl].

The receipt term itself is O(1) in history length: `:erts_debug.size` measures the sealed receipt at 40 words — 320 bytes on a 64-bit VM — at every scale from 10 to 100,000 leaves [pb4-merkle-dual.jsonl]. (The draft's "352-byte Compact Trust Receipt Token" figure has no measured anchor in this repository and is dropped per the number law; the measured 40-word term stands in its place.) Seal cost scales as measured: p50 0.024 ms at 10 leaves, 0.314 ms at 100, 64.7 ms at 10,000, 656.1 ms at 100,000 for the current sorted-tuple path, with a falsified O(n) variant honestly recorded as *not* an improvement [pb4-merkle-dual-path.md] — the dual-path bench is a refuted optimization, and the receipt says so.

Identity of artifacts across engines is anchored on canonicalization: the wasm binary is pinned by `priv/graphlaw.wasm.blake3` [gws11-wasm-memory-wall.md], and the cross-engine court reproduces the COLD-1 audit hash (`6971c912...`, sha256 over the canonical query and literals) byte-exactly in a second language, with the quarantined/released partition identical across native-ABI, wasm, and duckdb+N3 legs [gws12-cross-engine-determinism.md]. This is the repo's working instance of the W3C RDF Dataset Canonicalization (RDFC-1.0) claim model: hash the canonical form, seal the hash, and the proof travels across engines and languages. The admissibility argument is structural, not asserted: an immutable sealed segment with a sha256 manifest and last-written sentinel is the WORM-style record shape contemplated by FRE 902(14) (self-authenticating electronic records), FRCP 37(e) (sanctions curability where ESI was lost through routine good-faith operation), and

21 CFR Part 11 (audit trails and record-retention controls). The paper makes no legal conclusion; it exhibits the mechanism.

4 Measured results

Surface	Figure	Value	Artifact
Cosmos TLS point read	p99 / ops/s	0.543 ms / 2,251.4	`court_bench_cosmos.jsonl`
Cosmos TLS point read (run 2)	p99 / ops/s	0.672 ms / 2,130.0	`court_bench_cosmos.jsonl`
Native KV point read (etag/3)	p99 / ops/s	2 μs / 895,335	`powl-native-face.jsonl`
Native CAS put-if-match	p99 / ops/s	5 μs / 383,186	`powl-native-face.jsonl`
Wire-face point read	p99	0.813 ms	`powl-native-face.jsonl` (T43 comparison)
AMQP frame decode	throughput	952,381 frames/s	`court_bench_amqp.jsonl`
AMQP frame decode	cost	1.05 μs, 254.5 reds	`court_bench_amqp.jsonl`
4-step settlement (runner)	ops/s / p99	3,222.3 / 0.425 ms	`court_bench_powl.jsonl`
4-step settlement (runner)	reds/op	45,182.7	`court_bench_powl.jsonl`
4-step settlement (native)	p50 / p99	30 μs / 74 μs	`powl-native-face.jsonl`
Cross-language parity	ops/s	2,117.5 (erl) vs 11,426 (powl2)	`pb5-crosslang.jsonl`
Governance tax	median / reds	0.25 μs/step / +47.24	`gl9-governance-tax.jsonl`
Governance tax (tail)	guarded p99	0.976–1.147 ms	`gl9-governance-tax.md`
JSON SAX stream (50 MB)	wall / peak heap	80 ms / 38,504 B	`court_bench_json.jsonl`
Concurrency ceiling	μs/RT, reds/RT	0.9–1.18; 17.65→60.12	`court_bench_ceiling.jsonl`
Columnar projection	rows/s / RSS	450,450.5 / 100.0 MB	`sqld8-ingest-baseline.jsonl`
Dump converter	rows/s / RSS	64,724.9 / 294.2 MB	`sqld8-ingest-baseline.jsonl`
Merkle seal	p50 @10/100/10k/100k	0.024/0.314/64.7/656.1 ms	`pb4-merkle-dual.jsonl`
Receipt term size	erts_debug.size	40 words = 320 B, all n	`pb4-merkle-dual.jsonl`
Affidavit admit+seal	p50 / p99	11.265 ms / 32.671 ms	`c10-affidavit-seal.jsonl`
Crash-only recovery	cascade / drift	0–1 ms / +237 KiB/10 cyc	`ja7-crash-only-court.md`
WASM memory wall	caps	16 MiB req / 256 MiB budget	`gws11-wasm-memory-wall.md`
Release cold boot	best vs ceiling	1,906 ms vs 500 ms (FAIL)	`t47-release-boot-gate.md`
Ingest→settlement loop	assertions / leaves	112 / 47	`gld7-ingest-to-settlement-court.md`
Zero-row-loss round trip	rows	6/6, duckdb 1.4.1	`gld5fix-null-formula-court.md`
Cross-engine determinism	legs	3/3 byte-identical	`gws12-cross-engine-determinism.md`

Table 2: All measured results. Machine: Apple M3 Max, 16 cores, 48 GB, Darwin 26.2, OTP 28/29 per artifact (embedded per-row in each jsonl).

5 Macroeconomic implications

The draft's macroeconomic claims are kept qualitative and number-free.

Conway's Law inversion. Conway's Law says system structure mirrors organizational communication structure. The legacy stack therefore requires organizations to maintain communication paths (teams, contracts, meetings) for every runtime coordinate edge. Under lithography, coordinate edges are replaced by law edges: two components interact through admitted SHACL/POWL structure, not through organizational negotiation. The architecture no longer mirrors the org chart; the org chart becomes free to mirror the law graph.

Collapse of defensive scaffolding. The legacy stack's validators, retry storms, circuit breakers, and reconcilers exist because bytes arrive unproven. When admission is total at ingress and provenance is total at egress, the scaffolding has no job. What remains is measured in this repo: a guard seam costing 0.25 μs/step and +47.24 reds/step (GL-9), not an unmeasured framework tax. Crash-only semantics replace graceful-degradation machinery: recovery after a crash costs 0–1 ms of cascade and 237 KiB per ten cycles (JA-7), not a reconciliation subsystem.

Audit as byproduct. In the legacy stack audit is a project; in lithography the sealed receipt is emitted whether or not anyone asks. The admissibility surface (FRE 902(14)/FRCP 37(e)/Part 11) becomes a property of the write path rather than a compliance program, and cross-engine

byte-identity (GWS-12) makes the proof portable across implementations rather than owned by one vendor's stack.

6 Threats to validity

This section is the paper's honesty ledger; each item is a repo-witnessed limit.

- **Hot-upgrade is UNSUPPORTED.** Zero `.appup/.relup` artifacts and zero `code_change/3` across all 34 stateful processes; the release path is stop→rebuild→start ja14-release-handling.md. The (JA-12) court ja12-hot-reload-court.md proves only the OTP *mechanism* (dynamic module call, one-message lag), not a shipped upgrade capability.
- **Journal boot-replay is NOT WIRED.** The seal protocol is courted (SJ-1), but `JournalReader` has zero non-test callers; no boot-time replay into `Smalt.Store`, whose restart is documented lossy (`store.ex:220`); persistence is opt-in `DetsMirror` only ja14-release-handling.md.
- **WASM boundary is 14 operations.** The sandbox surface is bounded by WASM-SANDBOX-SPEC §5/§6; the memory wall is enforced at module caps (16 MiB request, 256 MiB outstanding) with the 64 MiB ceiling living at the *host* layer, not declared module memory gws11-wasm-memory-wall.md.
- **Store blast radius.** `Smalt.Store` is a single process on the hot path of every data-plane write; its crash fails every ARM/Cosmos/SB write for the restart window — the kernel is shallow but not minimal (JA-6) ja6-error-kernel.md.
- **Open defects JA-16 D1–D3.** D1 (P1): wire-sourced unbounded atom minting in `cosmos_sql.ex:766` — a DoS vector that crash-only cannot recover (the atom table is never GC'd). D2 (P2): silent CA-file load failure in `x_arr_cert.ex` — fail-closed but unobservable. D3 (P3): silent quota fallback in `eh_throttle.ex:243` ja16-robustness-census.md.
- **Bench-scale tails are noise-dominated.** At $\sim 6\,\mu$s absolute scale, guarded p99 (0.976–1.147 ms) is a scheduler/GC artifact; reductions, not timers, are the run-stable denominator (GL-9). Wire-vs-native gap decomposition is explicitly estimate-only powl-native-face.jsonl.
- **Cold start fails its own budget.** Release boot best 1,906 ms against a 500 ms ceiling — disclosed FAIL t47-release-boot-gate.md.
- **Draft figures with no anchor were dropped.** The $\leq 2.0\,\mu$s WAL append and 1.842 ms/10^7-row claims have no bench artifact here and do not appear; the "352-byte" token is replaced by the measured 320-byte receipt term; the "$<$ 0.2 ms Merkle verify" claim has no artifact and is replaced by the measured seal-cost ladder (PB-4).

References

[1] W3C. *RDF Dataset Canonicalization (RDFC-1.0)*, W3C Recommendation.

[2] W3C. *PROV-O: The PROV Ontology*, W3C Recommendation.

[3] van Dongen, Röglinger et al. *Object-Centric Event Data Structure (OCEL) 2.0 Specification.*

[4] W3C. *SHACL Shapes Constraint Language*, W3C Recommendation.

[5] Federal Rules of Evidence, Rule 902(14) (records generated by an electronic process that produces an accurate result, self-authenticating via certification).

[6] Federal Rules of Civil Procedure, Rule 37(e) (failure to preserve electronically stored information).

[7] 21 CFR Part 11 (Electronic Records; Electronic Signatures), U.S. FDA.

[8] Fahland, D. *Declarative, Object-Centric Process Representations: POWL* (SESE block structure); smalt GL charter receipts.

[9] O'Connor, L. et al. *BLAKE3: one function, fast in software.*

[10] Apache Arrow: cross-language development platform for in-memory columnar data.